

$$E=mc^2$$

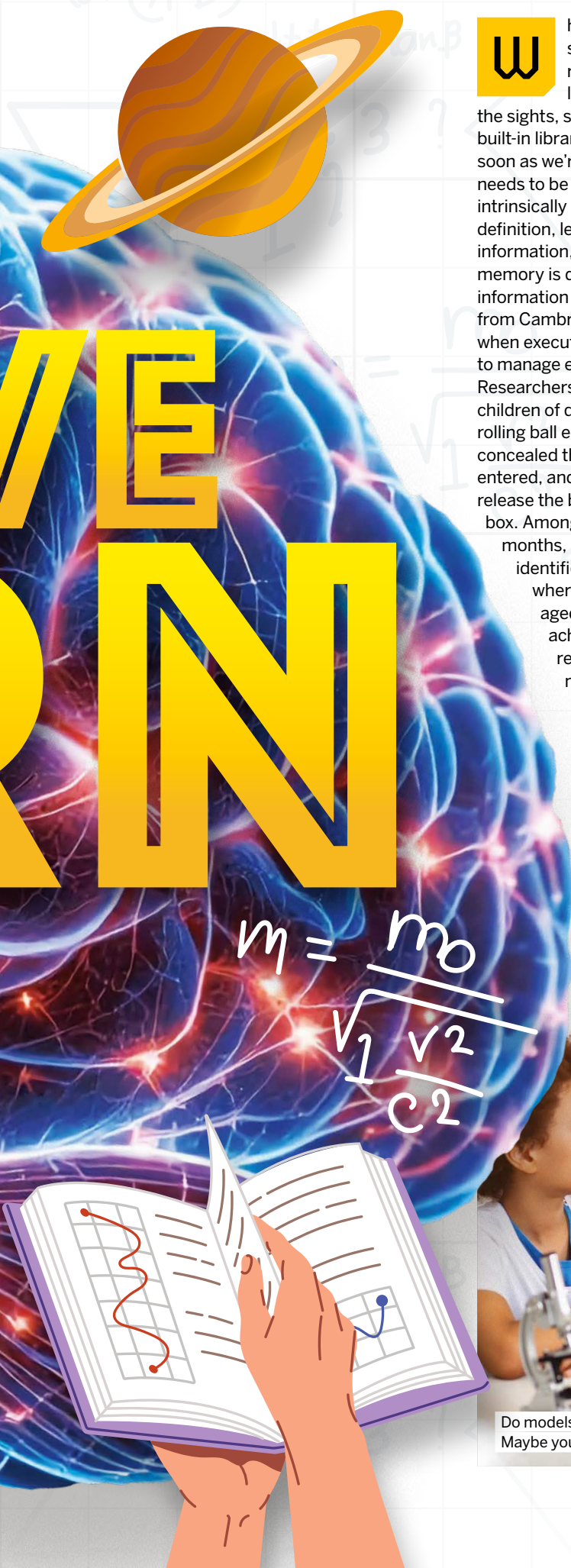
HOW WE LEARN

Every day is a lesson for everyone, though your brain will process and store your experiences differently to anyone else

WORDS SCOTT DUTFIELD

$$\sin^2 \alpha + \cos^2 \beta = 1$$

DID YOU KNOW? By the time a child reaches 18 years old, they will have spent nine per cent of their life in a classroom



Whether it's during the formative student years or a hobby-filled retirement, our brains are always learning new information. Fed by all the sights, sounds and smells around us, our built-in library of knowledge starts growing as soon as we're born. But how does it learn what needs to be learned? As a concept, learning is intrinsically intertwined with memory. By its definition, learning is the acquisition of new information, behaviours or abilities, whereas memory is defined by the ability to retain that information over time. In 2016, researchers from Cambridge University wanted to find out when executive skills (basic cognitive functions to manage everyday tasks) develop in humans. Researchers set up a series of tests involving children of different age groups watching a rolling ball enter one of four boxes, which concealed the ball once it had entered, and then asked them to release the ball from the correct box. Among those aged 29 to 31 months, 32 per cent correctly identified the ball's location, whereas 66 per cent of those aged 32 to 36 months achieved the same. The researchers concluded that a single month could make the difference in their ability to carry out the task. Our brain's ability to juggle the information it receives and store it as knowledge comes from a process called neuroplasticity. This adaptive ability lets cells in

the brain, called neurons, change their structure and connections with each other. For example, when you begin learning to ride a bike, your brain cells start to change. The more you practise pedalling, the closer the synaptic junctions – the connections – between some of these neurons become, as the neurons start branching out to other neurons to make more connections. Also, the electrical signals that travel along each of these neurons are strengthened. Over time, if you've been dedicated enough to master the skill of bike riding, you'll have formed neural connections that commit your newly gained ability to ride a bike to long-term memory.

The journey of learning, from first putting your hands on the bike's handlebars to becoming a self-assured cyclist, takes several stages. First, the brain has to select what information is important enough to continue through the cerebral processing plant for eventual storage, while the rest is quickly lost. The information the brain deems worth remembering is a complex process, and science still isn't certain on the specifics. The emotional importance of information, such as the desire to ride a bike or if there's a reward associated with developing the skill, has been associated with what the brain chooses to remember.

In 2024, researchers published a study in the journal *Science* that suggested the brain may 'tag' experiences worth remembering by repeatedly sending out high-frequency brain

DID YOU KNOW?

There are 100 trillion neural connections in the brain



Do models help you learn a new concept? Maybe you're a kinesthetic learner

An illustration of a synaptic junction between neurons

IS AI MAKING YOU DUMBER?

According to a recent study by the Massachusetts Institute of Technology, it might be. In 2025, researchers split a group of 54 people, aged 18 to 39, into three groups and asked them to write several Scholastic Aptitude Test (SAT) essays. The first group was asked to do so with the help of OpenAI's ChatGPT, the second used Google's search engine and the third used nothing. While the subjects were busy working, the researchers used an electroencephalogram (EEG) to record the electrical activity of 32 different regions of the brain. The results of the study found that over several months of observations, the ChatGPT users underperformed in several areas and had lower cognitive engagement when compared to the other two groups. The researchers concluded that "while AI tools can enhance productivity, they may also promote a form of 'metacognitive laziness' where students offload cognitive and metacognitive responsibilities to the AI, potentially hindering their ability to self-regulate and engage deeply with the learning material."

"This consolidation of neurons is what forms the knowledge library of your brain"

waves, known as 'sharp wave ripples'. Researchers monitored the brain activity of mice moving through a maze, along with any ripples, while they slept. They found that the ripples replayed the patterns of brain activity while they slept, as well as the ones that occurred while the mice took breaks to rest or groom during the maze run. The researchers suggested that the resting ripples might act as a tag to consolidate memories later on while they sleep.

Once the brain has selected what it wants to retain, it processes it in our 'working memory'. Also known as short-term memory, the brain temporarily stores information you've learnt, such as someone's telephone number, for the sake of completing a task, like making a phone call. However, for retaining knowledge, temporary information in our working memory needs to be transferred into our long-term memory through an initial process called encoding. Encoding is like the brain's built-in scribe, recording the sensory information to be filed away in the library of your long-term memory. Once

encoded, the information is then strengthened in order to be shelved in your brain library through the process of consolidation. This is achieved largely through being repeatedly exposed to the information.

Why do you think you know how to say "my name is..." in French, when the Battle of Hastings took place or the entire lyrics of Queen's *Bohemian Rhapsody*? It's because you were made to say "je m'appelle" repeatedly in school, you heard the incessant jingle of an insurance advert in the 2000s or because the sound of Freddie Mercury's stellar voice frequently plays on the radio. Consolidation is also when the previously mentioned neuroplasticity of the brain comes into play and synaptic junctions strengthen. This consolidation of neurons is what forms the knowledge library of your brain, and the process of checking out those books of knowledge is known as retrieval.

Without the late fees or borrowing limits of a standard library, you can retrieve this stored knowledge. However, it's not a playback of exactly what was initially recorded: it's a reconstruction that can sometimes be

DID YOU KNOW?

It's a myth that the prefrontal cortex isn't fully developed until the age of 25

DID YOU KNOW?

60 per cent of the human brain is made of fat

unknowingly modified or distorted, depending on the strength of the consolidation.

As we all know, no two people are the same, and neither are their brains. A combination of genetics and environmental factors, such as nutrition and education, means that people learn in different ways and at different speeds. And according to American psychologist Dr Carol Dweck, our mindsets can even impact how well we learn. In 2006, Dweck developed the concept of 'fixed and growth mindsets'. She suggested that people have either a fixed mindset, where intelligence and ability are innate or fixed, or a growth mindset, which allows for intelligence and ability to develop over time.

Having a growth mindset alone might not be enough to help you understand how you learn. Knowing what type of learner you are can also massively improve your cognitive function. Do you learn better from listening to a lecture, analysing a diagram or reading a textbook? In 1992, Neil Fleming and Colleen Mills created a tool for students to help them discover their sensory preferences when it comes to learning, known as the VARK model: Visual, Aural, Read/Write and Kinesthetic. While the first three are self-explanatory, the last letter in the acronym refers to those who prefer a practical element

1 NEURONS

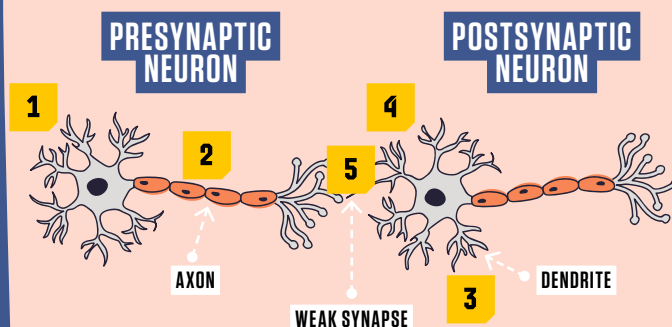
A neural connection involves neuron cells: presynaptic neurons, which transmit a signal, and postsynaptic neurons, which receive it.

CLOSING THE GAP

What neurons look like when we're in the process of learning

2 AXON

The electrical signals involved in neuron communication travel along part of the neuron called the axon.

NEW SKILL / INFREQUENT ACTIVITY**3 DENDRITE**

The branch-like protrusions are dendrites, which receive the electrical signal by way of neurotransmitters.

4 COMMUNICATION

To carry the electrical signal from one cell to another, the presynaptic neuron releases chemicals called neurotransmitters across the gap between the cells, called the synaptic junction.

5 WEAK SYNAPSE

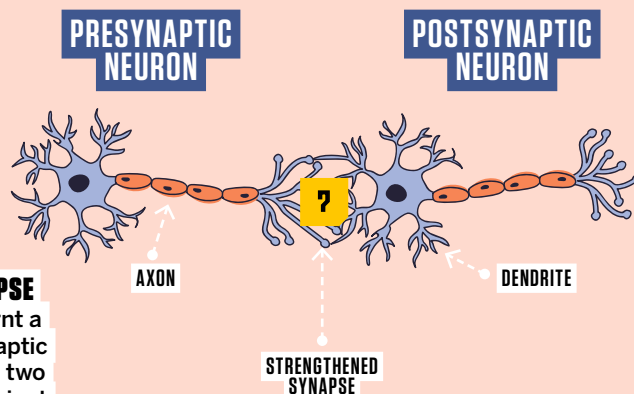
Before you've learnt a piece of information, the synaptic junction is weak, with a lower release and uptake of neurotransmitters.

REPETITION
AND
LEARNING

6

6 REPETITION

The repeated learning of a new skill or piece of information strengthens the synaptic junction.

PRACTISED SKILL / REPEATED ACTIVITY**7 STRONG SYNAPSE**

When you've learnt a new skill, the synaptic junction between two cells is highly efficient and the cells are in close proximity.

to learning, such as demonstration or simulations. People can fall into multiple learning categories, but the tool helps identify which method of learning might be best suited to different people.

Between 2022 and 2023, VARK Learn Limited, the company that owns and manages the VARK model, surveyed more than a million people who completed the Standard VARK Questionnaire. The results found that just 34 per cent of participants had a single preference when it came to learning styles, with the remainder having a 'multimodal preference', combining styles. Among the single preferences, kinesthetic learners were the most common, at around 23.3 per cent.

Our ability to learn new information is defined by our intelligence. Generally speaking, the more intelligent a person is, the greater their learning capacity. Combining all areas of our cognitive functions, including problem-solving, reasoning and learned knowledge, intelligence refers to the ability to adapt to new situations and apply knowledge to our environments. We typically measure our intelligence using an IQ (intelligence quotient) test. First introduced in 1904 by Alfred Binet, a French psychologist, and his psychiatrist partner Théodore Simon, in order to evaluate a child's need for special education, the IQ test is the best known indicator of human intelligence around the world. With more than a century of development, today's IQ tests have been designed by psychologists to evaluate different aspects of our cognitive



The brain's frontal lobe regulates decision-making and problem-solving

DID YOU KNOW?

The average human brain weighs 1,300 to 1,400 grams

HOW TO IMPROVE YOUR SMARTS

Here's some top tips to help you get the most out of your study session



GET A GOOD NIGHT'S SLEEP

Studies have found that sleeping helps us reorganise recently learned information and improve memory and recall. Researchers analysing the Adolescent Brain Cognitive Development (ABCD) Study, the largest long-term study of brain development in the US, found that children who slept the longest on average (7 hours and 25 minutes) performed better in cognitive function than the other test group, who slept for less time.



CHOOSE THE RIGHT PLAYLIST

Listening to the right music while studying can improve your learning ability. Online university CSU Global surveyed 2,000 US students and found that those who listened to music while studying were more likely to have a slightly higher Grade Point Average (GPA). Around 60 per cent of the surveyed students also said they were able to study better with background sound.



FIND THE RIGHT ENVIRONMENT

Finding an environment that suits your study style is crucial to learning. According to University College London, when it comes to classroom design, daylight had the biggest impact on student progress and supported academic performance. Some studies have shown that students who studied in naturally lit classrooms achieved up to 25 per cent higher scores in standardised tests than other students.

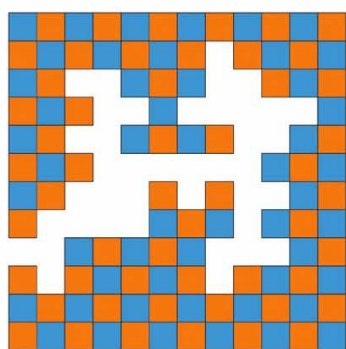
DID YOU KNOW? Mensa, the IQ society, accepts members who score in the top two per cent of the population

functions, including logic, pattern recognition and working memory.

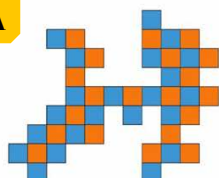
In 2024, Dr YoungHoon Kim claimed the world record for the highest IQ score at 276. For comparison, one of the greatest known minds, Albert Einstein, is estimated to have been around 160. The average IQ is 100, but it isn't stagnant. It's often assumed that we're born with a certain level of intelligence and would therefore score a certain way on an IQ

test. However, our intelligence and subsequent IQ score can change over time. Some studies have estimated that our fluid intelligence, which is the ability to think abstractly and problem-solve, peaks at around 20 to 24 years old and then slightly declines as we age. Similarly, our crystallised intelligence – the ability to utilise acquired skills through education and experience – might peak at around 35 to 44 years old before declining as well.

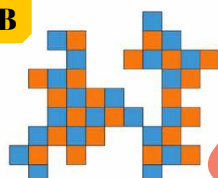
IQ test questions are designed to assess a person's cognitive function, including spatial recognition. Which shape fits?



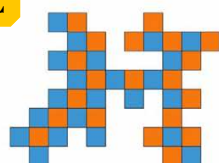
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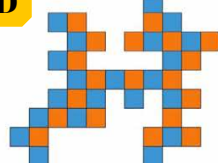
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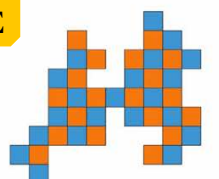
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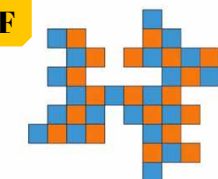
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IS 10,000 HOURS ENOUGH?

You might have heard the phrase "it takes 10,000 hours to master something." But is that true? According to the BBC, the concept of a 10,000-hour quota for mastering a skill can be traced back to a paper written by K. Anders Ericsson, a professor at the University of Colorado, in 1993, entitled *The Role of Deliberate Practice in the Acquisition of Expert Performance*. Ericsson cited the work of a group of researchers in Berlin, Germany, who monitored a group of violinists from the age of 5 to 20 years old. Those who had practised for more than 10,000 hours had become classed as expert performers.

However, this golden time frame for learning was popularised by Canadian journalist Malcolm Gladwell in his 2008 book *Outliers*, which Ericsson publicly disputed. Ericsson wrote a rebuttal to the book in 2012, stating that the 10,000 hours was an average figure of the study. "Most of the best musicians had accumulated substantially fewer hours of practice at age 20," he wrote. So if you're thinking of picking up the violin or learning a new language, don't worry, there's still plenty of time to learn.



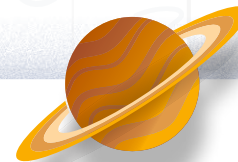
GET SMART SNACKS

Maintaining a healthy and balanced diet, unsurprisingly, has a positive impact on our brains' overall health and function. Some studies have identified some legitimate 'brain foods' to support studying, including those with an abundance of omega-3 fatty acids, such as salmon and walnuts, along with some berries packed with flavonoids, such as blueberries, which can help improve memory.



TURN OFF NOTIFICATIONS

According to the Pew Research Center, around 95 per cent of US teens have access to a smartphone. While they can be a great way to access information, they're also an addictive distraction. Setting your phone to aeroplane mode or limiting notifications can help focus your attention on studying, as well as installing productivity tools such as the Flora app.


DID YOU KNOW?

The hippocampus can store up to 4.6 bits of information

WHAT TYPE OF LEARNER ARE YOU?

Answer these simple questions to find out what your learning style might be

You've bought a piece of flat-pack furniture. How do you start the assembly?

- A** ☐ By looking through the diagrams in the instruction manual
- B** ☐ Have someone else read through the instructions while you build
- C** ☐ Watch a video tutorial
- D** ☐ Read through all of the assembly instructions

You're walking through a museum. How would you prefer to receive information about an artefact?

- A** ☐ Illustrations of its origin, including maps of where it was found
- B** ☐ Ask a museum staff member to tell you about the artefact
- C** ☐ Engage with props or models that illustrate the artefact's history
- D** ☐ A written description of its history

There's an issue with your elbow. How would you like the doctor to explain it?

- A** ☐ With an anatomy diagram of the elbow
- B** ☐ Verbally explaining the issue
- C** ☐ With a model of the elbow to demonstrate the issue
- D** ☐ Using some literature on the issue for you to read through

When choosing a new laptop to purchase, what helps you make a decision?

- A** ☐ An infographic of each laptop's features and functions
- B** ☐ Talking to a sales assistant in a store for advice
- C** ☐ Trying out a display model in store
- D** ☐ Reading through the laptop's specifications and reviews

You're hosting a dinner party and you're looking for the right recipe. What helps you decide?

- A** ☐ Flicking through the photographs in a cookbook
- B** ☐ Calling up family members for inspiration
- C** ☐ Trying out a few different types of dishes beforehand to see what works for you
- D** ☐ Reading through food reviews and suggestions online

You've been put in charge of organising a corporate away day. How do you communicate the planned events?

- A** ☐ By sharing diagrams and maps of the planned schedule
- B** ☐ Arranging a conference call to discuss the away day
- C** ☐ Putting together a presentation for the away day itinerary
- D** ☐ By sending a detailed itinerary via email

For your first time in the gym, how do you get to grips with the equipment?

- A** ☐ Study the diagrams on each machine to understand how to use them
- B** ☐ Take a friend with the know-how to explain what to do
- C** ☐ Watch YouTube videos to see how others use the machines
- D** ☐ Read through online advice for using the equipment

You're asked for directions. How do you explain where they need to go?

- A** ☐ Draw a map of the roads ahead
- B** ☐ Verbally describe the best route
- C** ☐ Offer to guide them to their destination
- D** ☐ Write down a set of instructions

Answers key



A VISUAL You might be a visual learner. You prefer to see visual representations of information, also known as spatial learning. Graphs, diagrams and infographics are your studying best friend.

B AUDITORY You might be an auditory learner. Learning for you may be reinforced by sound. Rather than just taking study notes, hearing concepts and ideas helps you learn new information.

C KINESTHETIC You might be a kinesthetic or tactile learner. You prefer to physically interact with objects and models to understand new concepts and ideas.

D READ/WRITE You might be a reading and writing learner. The written word might be the best way for you to understand new information. Surfing the internet and reading text books is your bread and butter.